

《Automotive Configuration》

Lecturer : Dr. HU Zongjie/Associate Professor

Department: School of Automotive Studies,
Tongji University

Contact: zongjie-hu@tongji.edu.cn

Lecture Schedule: 2nd Semester Week 5~6

第五章 柴油机燃油供给系统

Chapter 5 Fuel Supply System for Diesel Engine



Overview

5.1 Characteristics and Properties for Diesel Fuel

柴油及其使用性能

5.2 Composition of Fuel Supply System for Diesel Engine

柴油机供给系统组成

5.3 Injectors

喷油器

5.4 Plunger Injection Pump

柱塞式喷油泵

5.5 Distributor Injection Pump

分配式喷油泵

5.6 Speed Governor

调速器

5.7 Electronic-controlled Injection System for Diesel Engine

柴油机电控燃油喷射系统

5.8 Auxiliary Device

电控燃油喷射系统辅助装置

5.1 Characteristics and Properties for Diesel Fuel (柴油及其使用性能)

1. Ignition Quality

——Cetane Number (CN) (自燃性的指标——十六烷值 $C_{16}H_{34}$)

CN $\uparrow$ →Ignition Quality $\uparrow$ →Ignition Delay (着火延迟时期) $\downarrow$ →Starting and combustion behavior(冷起动及燃烧特性) $\uparrow$ →Exhaust and noise emissions (排放及噪音) $\downarrow$

CN $\uparrow\uparrow$ →Molecular Weight(燃料分子量) $\rightarrow$ Evaporability(蒸发性) $\downarrow$
 $\rightarrow$ Viscosity(粘度) $\uparrow$ $\rightarrow$ Smoke Formation $\uparrow$ &Fuel Economy(燃油经济性) $\downarrow$

Cetane Number ≥ 45 (Presently lies between 51 and 56)

2.Evaporability

—Distillation(馏程): Temperature range where diesel evaporates to a specific percentage

•50% boiling point(50%馏出温度)presents the average evaporability(平均蒸发性能):

50% point ↓ →quick evaporation →mixture formation ↑; however, mixture formation gets too fast if 50% point is too low and would cause rough running of the engine.

•90%&95% boiling point(Final boiling point) presents the amount of heavy fraction which is hard to evaporate. Higher final boiling point would enlarge the droplets in the injection jet, and has negative influence on the combustion process.(90%及95%馏出温度表示难以蒸发的重馏分，偏高可导致喷雾蒸发困难，进而影响燃烧过程)

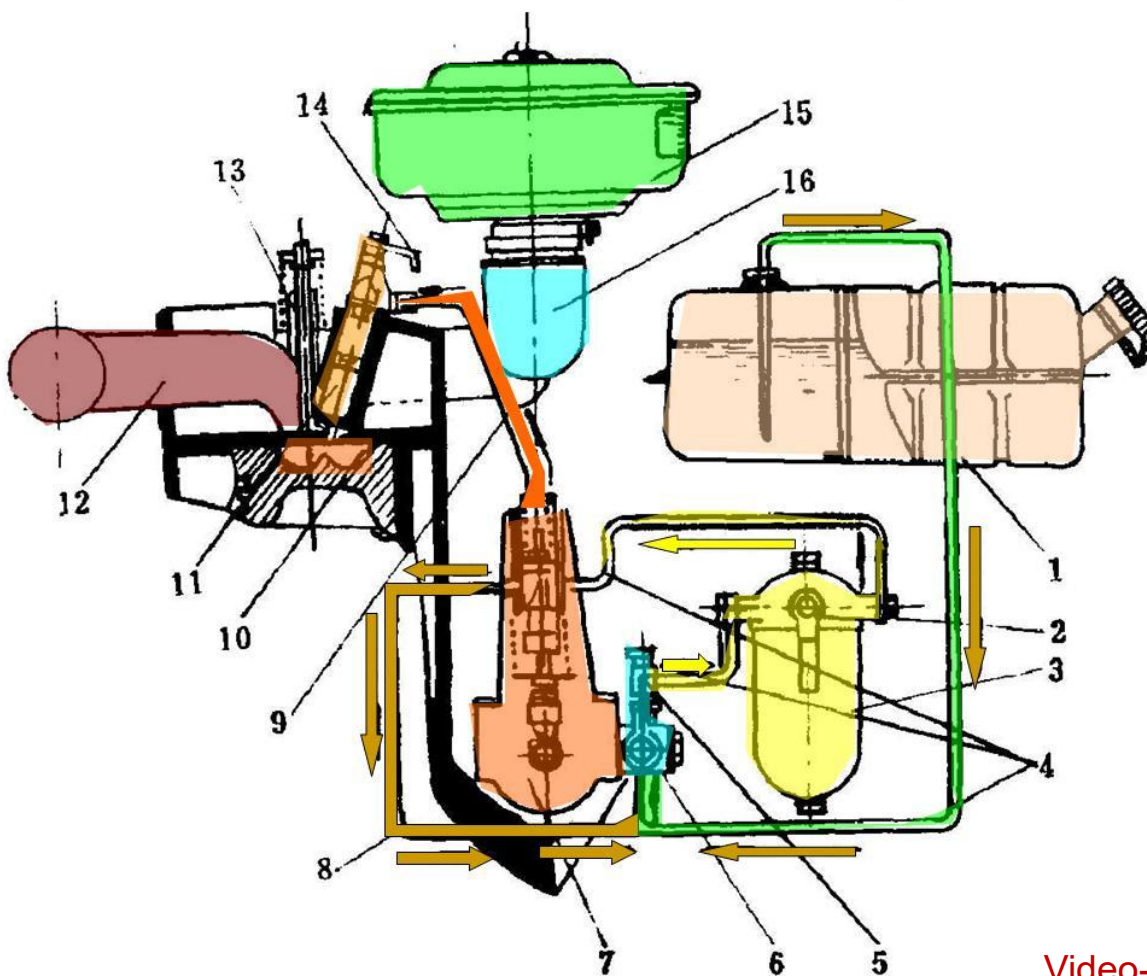
3. Atomization Quality(雾化性)—Viscosity(粘度)

- Too low: insufficient lubrication of the fuel injection system
- Too high: larger droplet size, poorer mixture formation, lower performance, higher soot emissions
- Kinematic viscosity(运动粘度): $(2.5 \sim 8) \times 10^{-6} \text{ m}^2 / \text{s}$ at 20°C

4. Flow ability(流动性)—Freezing Point(凝点)

- $T_{\downarrow}$ --cloud point(浊点)—lose flow ability
- Diesel classification: 10、5、0、-10、-20、-35 and -50

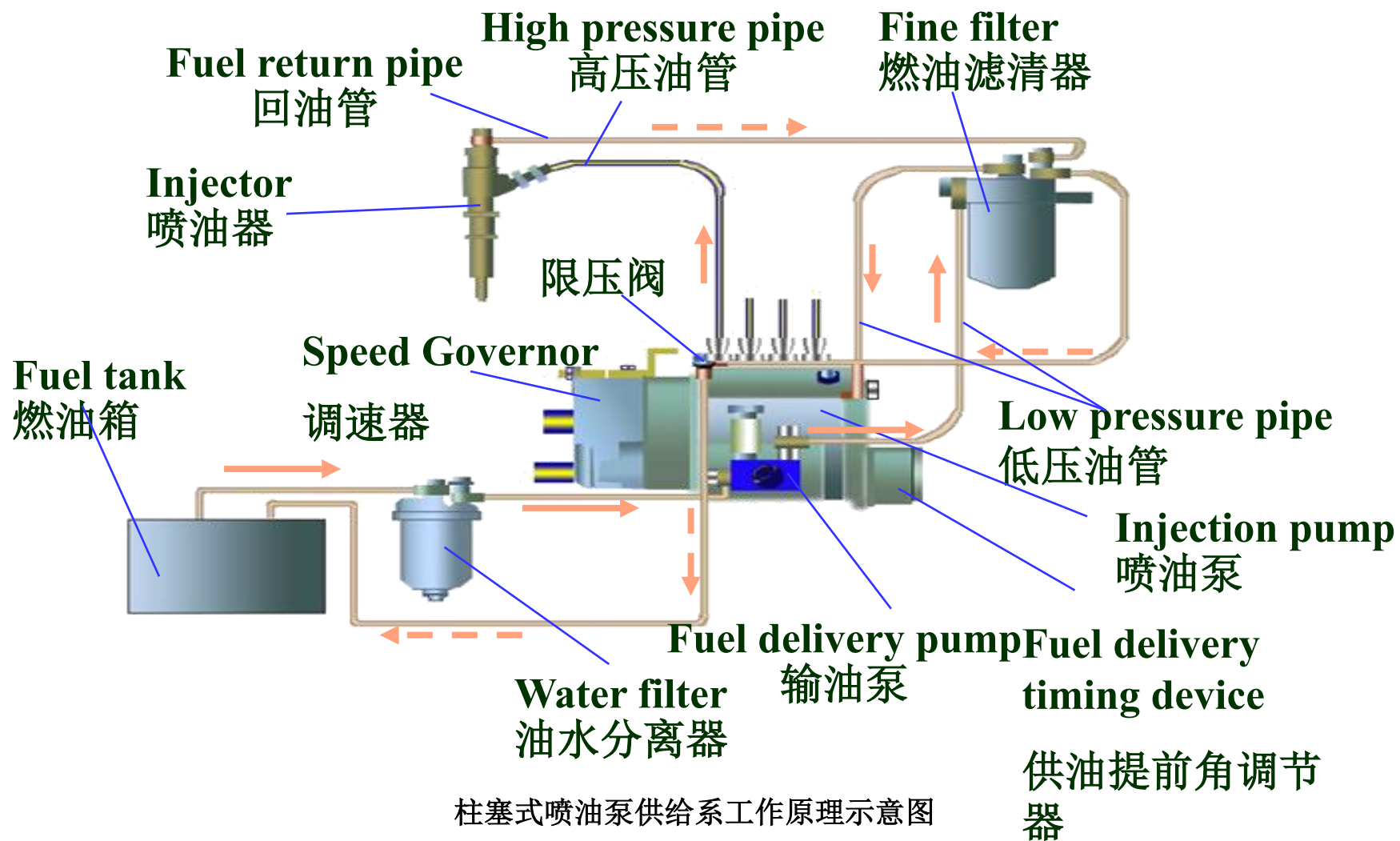
5.2 Composition of Fuel Supply System for Diesel Engine (柴油机供给系统组成)

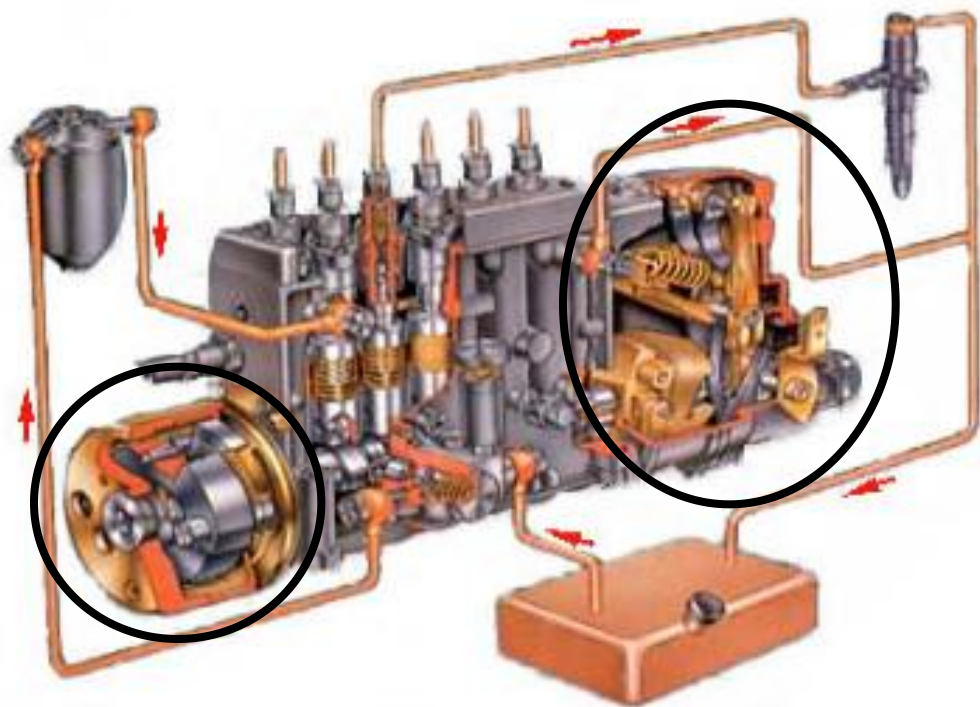


- 1- Fuel tank
- 2- Overflow valve
- 3- Fuel Filter
- 4- Low pressure pipe
- 5- Manual fuel supply pump
- 6- Auto fuel supply pump
- 7- Fuel injection pump
- 8- Fuel return pipe
- 9- High-pressure fuel pipe
- 10- Combustion Chamber
- 11- Fuel injector
- 12- Scavenge pipe
- 13- Exhaust valve
- 14- Fuel drain
- 15- Air filter
- 16- Intake manifold

- 1-柴油箱;
- 2-溢油阀;
- 3-滤清器;
- 4-低压油管;
- 5-手动输油泵;
- 6-机动输油泵;
- 7-喷油泵;
- 8-回油管;
- 9-高压油管;
- 10-燃烧室;
- 11-喷油器;
- 12-排气管;
- 13-排气门;
- 14-泄油管;
- 15-空气滤清器;
- 16-进气管;

Video-1

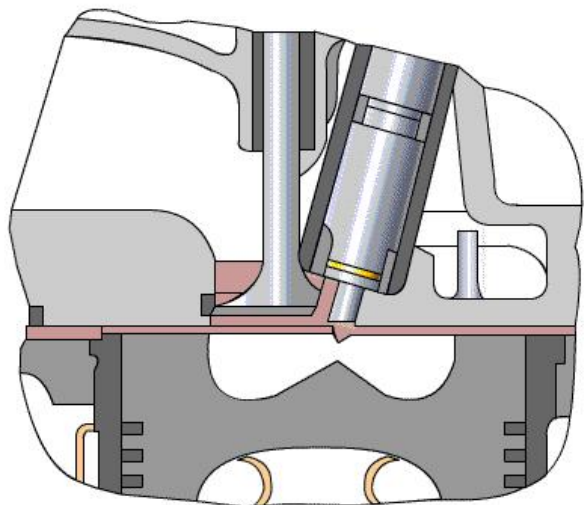




柱塞式喷油泵柴油机燃油系统示意图



5.3 Mixture formation and Combustion Chamber (混合气形成及燃烧室)



ω型燃烧室

Open combustion Chamber
(统一式燃烧室)

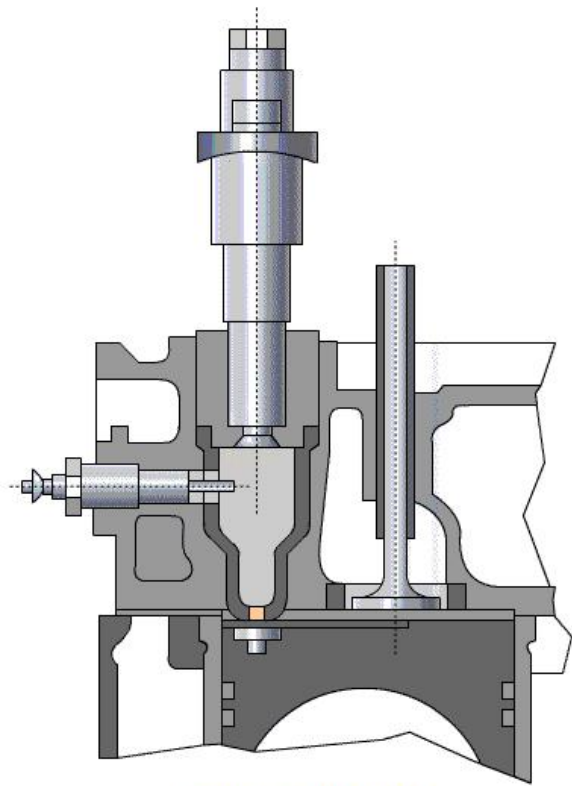
ω Type (ω型)

Spherical (球型)

Open Combustion Chamber

(统一式燃烧室)

- The actual combustion chamber is formed by recess in the piston head and bottom of cylinder head
- 单纯由活塞顶的凹面与气缸盖下底面所形成的燃烧空间这种形式的燃烧室燃烧效率高，热损失较少；但要求喷油压力高，油品质量较好。一般分为球型燃烧室及ω型燃烧室。



预燃室式燃烧室

Divided Combustion Chamber

(分隔式燃烧室)

- Combustion chamber consists of a main combustion chamber and a secondary combustion chamber.
- Main combustion chamber is located in piston crown, while secondary combustion chamber is located in cylinder head
- Fuel is injected into secondary combustion chamber first, and a pre-combustion process is carried out. After that, the mixture is pushed back into the main combustion chamber to finish the whole combustion process

Whirl combustion chamber

涡流室燃烧室

Pre-combustion chamber

预燃式燃烧室

Divided Combustion Chamber

(分隔式燃烧室)

5.4 Fuel Injector (喷油器)

- **Function:** Inject the fuel provided by injection pump into the cylinder with fine atomization and distribution

- **Basic design:** Pintle type, Multi-hole type

- **Requirements:**

- ✓ Appropriate spray corn angle (喷雾锥角) with fine atomization

- ✓ Injection with specific pressure and spray penetration(贯穿距)

- ✓ Quick cut-off without droplet leakage

- ✓ Appropriate spray shape and direction matched with combustion chamber

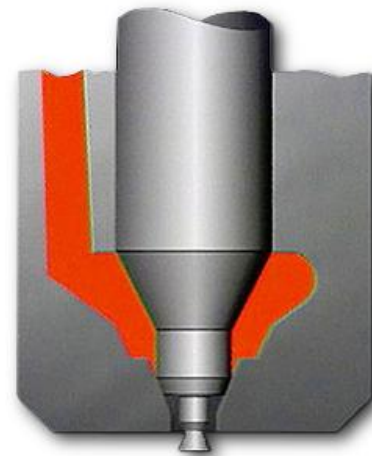
1、功用：将喷油泵供给的高压柴油，以一定的压力，呈雾状喷入燃烧室。

2、分类： 多孔式和轴针式

3、要求： ①合适的喷注锥角，良好的雾化质量； ②有一定的喷射压力和射程；
③喷油終了迅速停油，无滴漏现象； ④油束形状与方向，适应燃烧室。

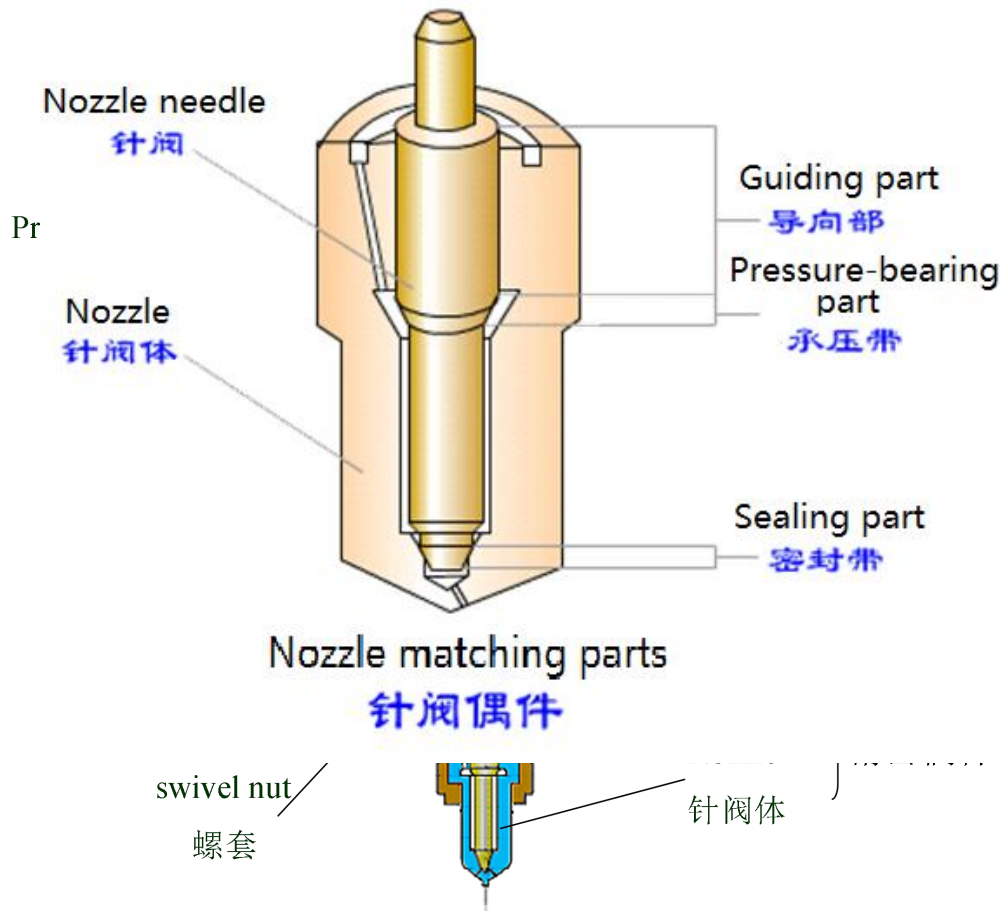


多孔式



轴针式

Structure of hole-type injector

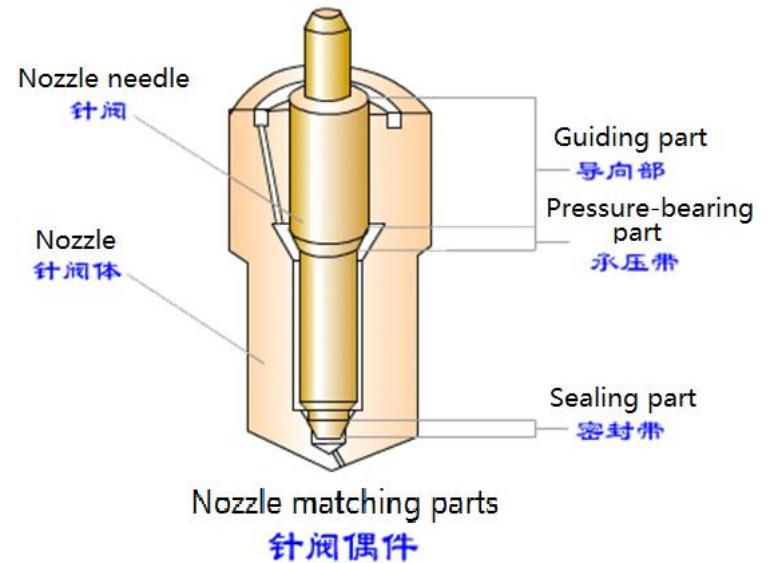
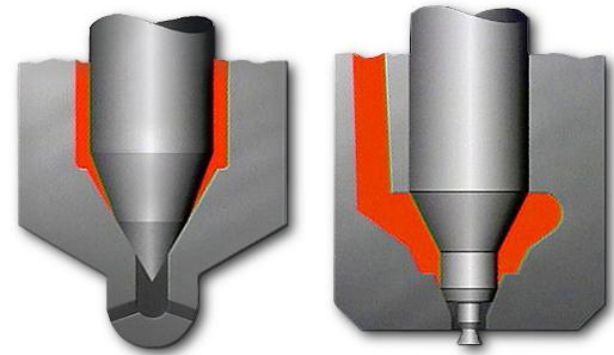
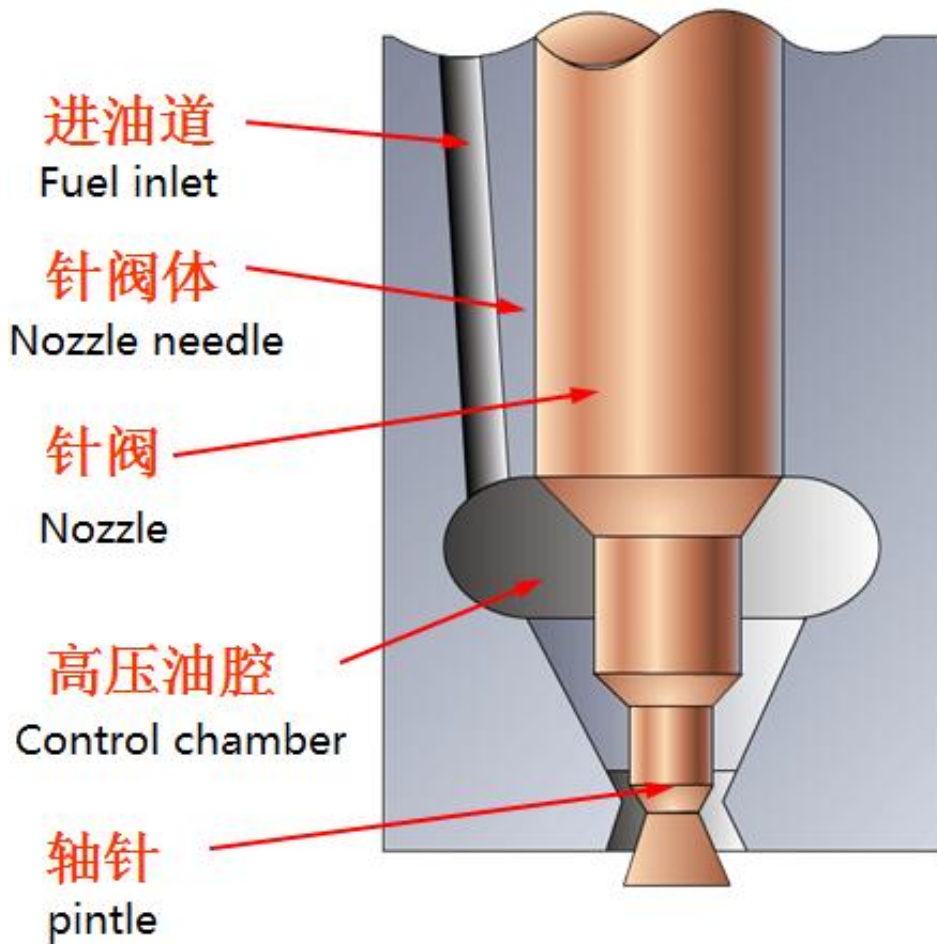


① Injection: fuel in high-pressure chamber forces the nozzle needle to move upwards → sealing cone departs from seat → injection

② Cut-off: Fuel pressure drops → pressure adjusting spring forces the needle to move downwards → sealing cone back → nozzle closed by needle

③ Fuel return: Part of high-pressure fuel returns to filter through gap between needle valve couple, lubricating and cooling the needle valve couple

Structure of Pintle-type injector



5.5 High-pressure Pump (喷油泵)

- **Function**

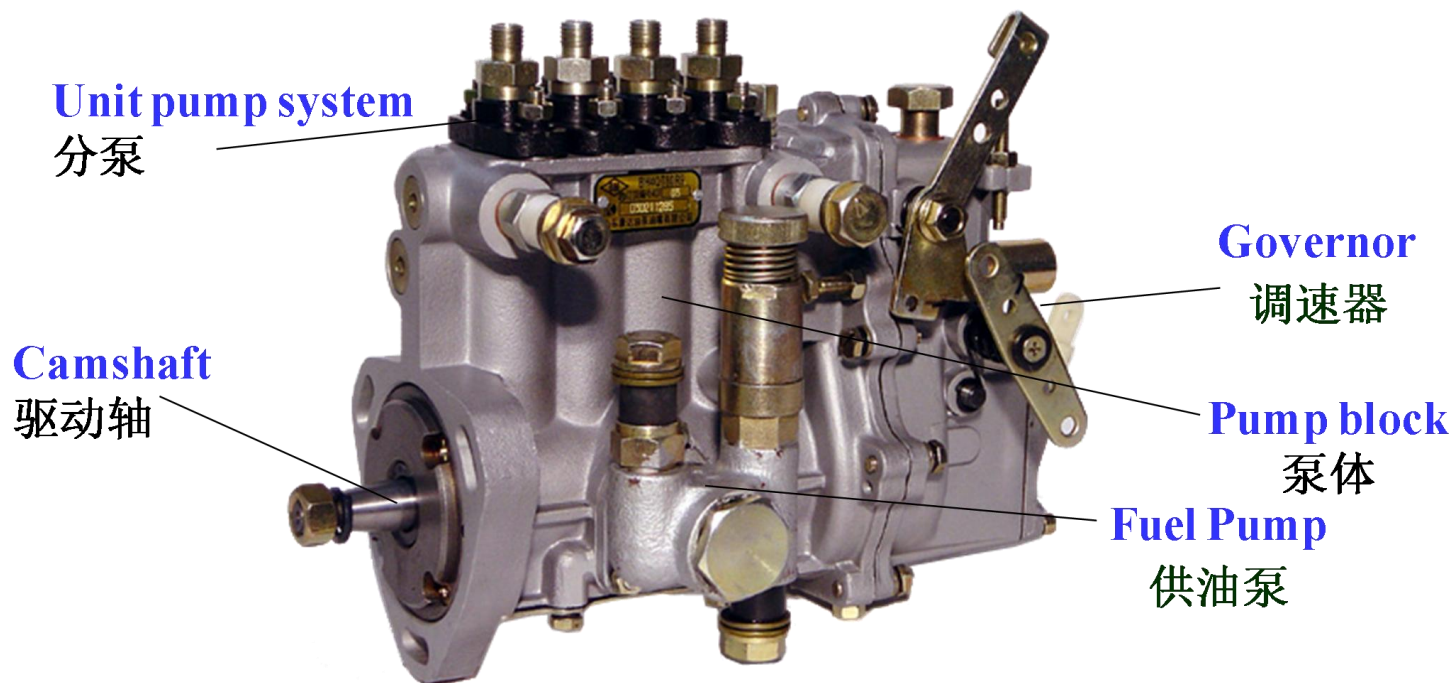
Deliver high-pressure fuel to injector according to engine needs, including injection order, engine load, injection timing, injection pressure, and injection mass

(按照发动机的工作顺序, 负荷大小, 定时、定量、定压地向喷油器输送高压柴油。)

- **Classification**

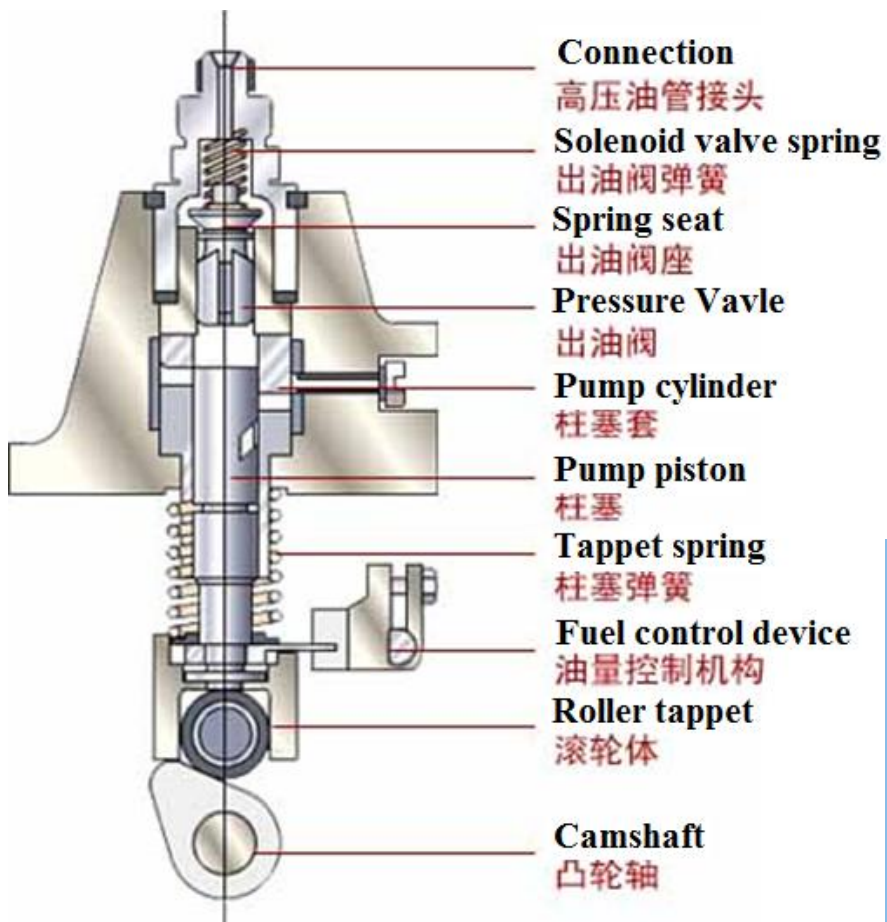
- ✓ Piston pump ①柱塞式喷油泵 (A、B、P、Z型)
- ✓ Distributor Injection pump ②转子分配式喷油泵 (VE)
- ✓ Pump-Nozzle System ③喷油泵- 喷油器 (单体泵或泵喷嘴)

In-line Fuel injection Pumps (直列柱塞式喷油泵)



- **Pumping mechanism**
(泵油机构)
- **Fuel control mechanism**
(油量调节机构)

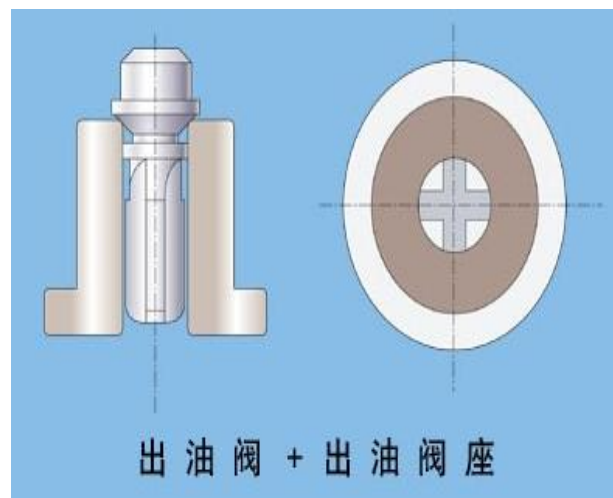
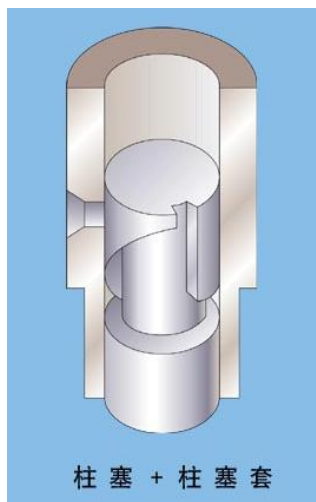
- **Driving mechanism**
(驱动机构)
- **Pump block**
(油泵体)

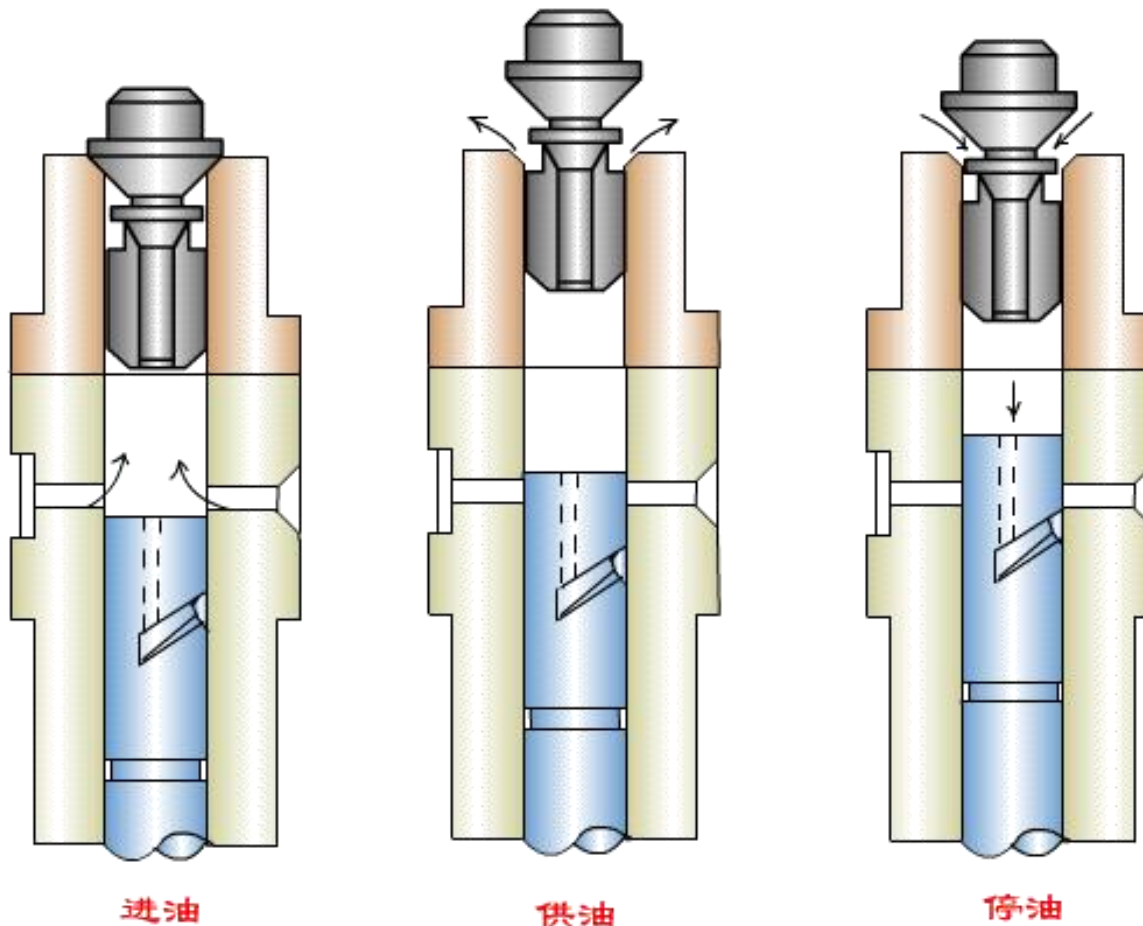


✓ **Plunger and barrel assembly**
 柱塞偶件

✓ **Solenoid Valve Couple**
 出油阀偶件

✓ **Roller assembly**
 滚轮体组





柱塞泵的泵油过程

Fuel inlet process; Fuel delivery process; Fuel return process

5.6 Speed Governor (调速器)

- **Function:** Stabilize engine speed under different engine loads by regulating fuel delivery

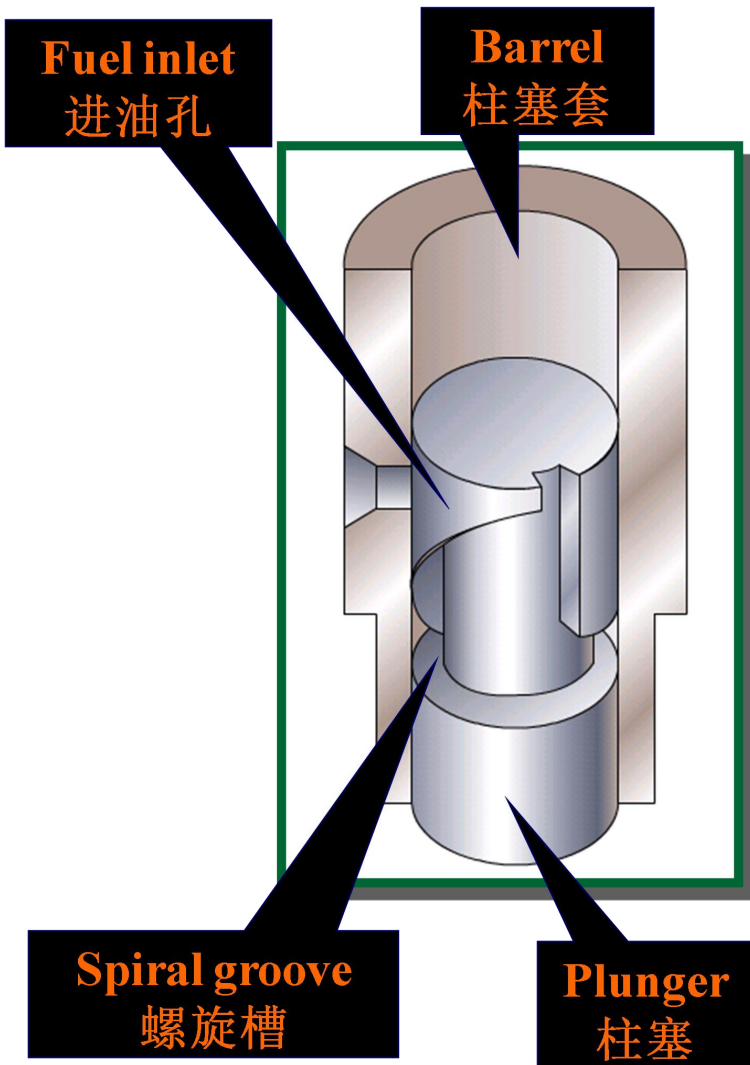
调速器是根据发动机负荷变化而自动调节供油量，从而保证发动机的转速在很小的范围内变化。

- **Speed performance of pumps:**

Engine speed $\uparrow$ $\left\{ \begin{array}{l} \text{Throttling effect } \uparrow \\ \text{Fuel leakage } \downarrow \end{array} \right\} \left\{ \begin{array}{l} \text{Start timing } \uparrow \\ \text{Cut off timing } \downarrow \end{array} \right\} \text{Fuel delivery } \uparrow$

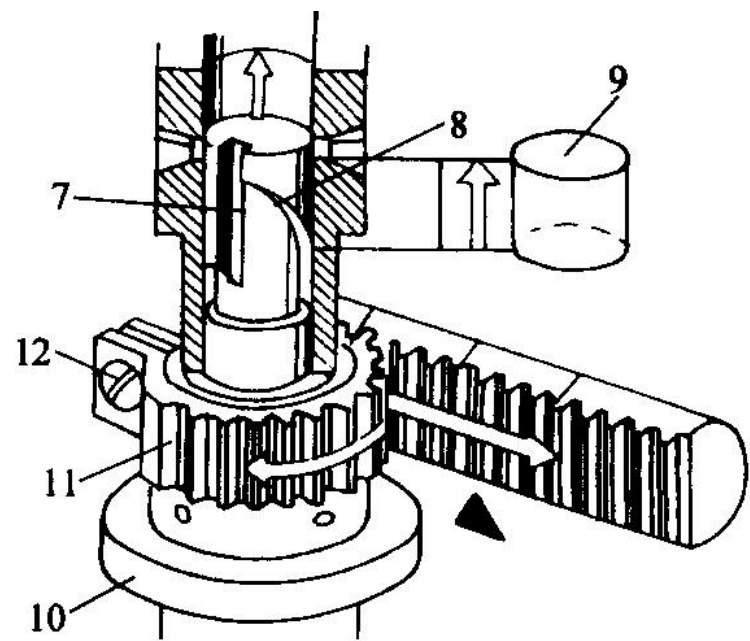
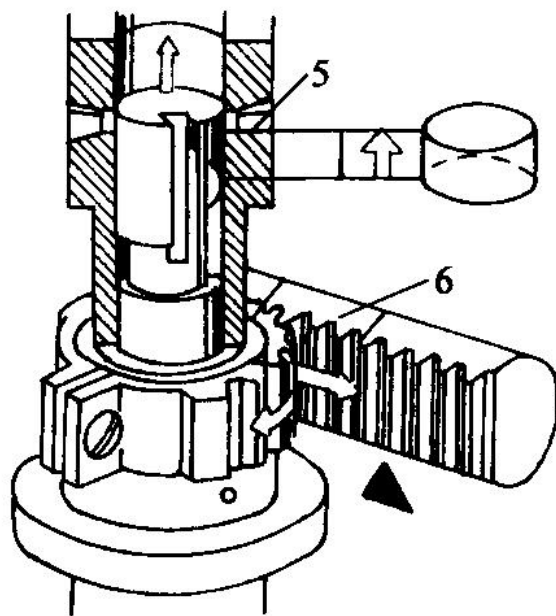
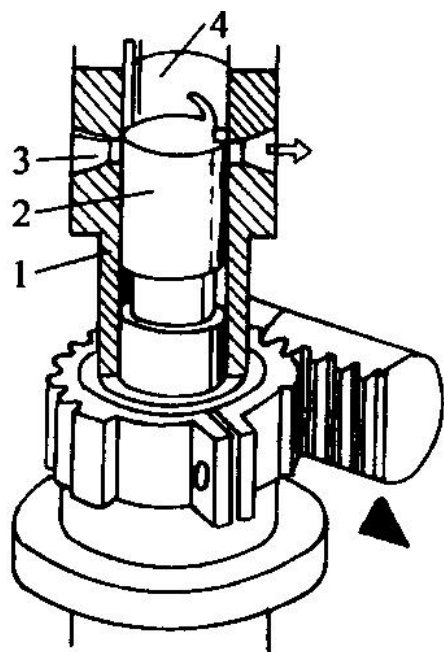
Engine speed $\downarrow$ $\left\{ \begin{array}{l} \text{Throttling effect } \downarrow \\ \text{Fuel leakage } \uparrow \end{array} \right\} \left\{ \begin{array}{l} \text{Start timing } \downarrow \\ \text{Cut off timing } \uparrow \end{array} \right\} \text{Fuel delivery } \downarrow$

Effect: unsteady engine speed $\left\{ \begin{array}{l} \text{High engine speed: } \underline{\text{overspeed (飞车)}} \\ \text{Idling: } \underline{\text{stop (熄火)}} \end{array} \right.$



• Plunger and barrel assembly

- ✓ A pair of precise coupled device, not interchangeable after grinding with each other
 - ✓ High precision and fineness with fit clearance of 0.001~0.003mm
 - ✓ Head of plunger is machined with spiral groove to control injection mass
- ▶ **柱塞和柱塞套**是一对**精密偶件**，经配对研磨后不能互换，要求有高的精度和光洁度和好的耐磨性，其配合间隙为0.001~0.003mm。
- ▶ 柱塞头部圆柱面上切有**斜槽**，并通过径向孔、轴向孔与顶部相通，其目的是改变循环供油量



- **Classification:**

- ✓ In terms of function: two-speed governor, full-speed governor, fixed-speed governor, combined governor

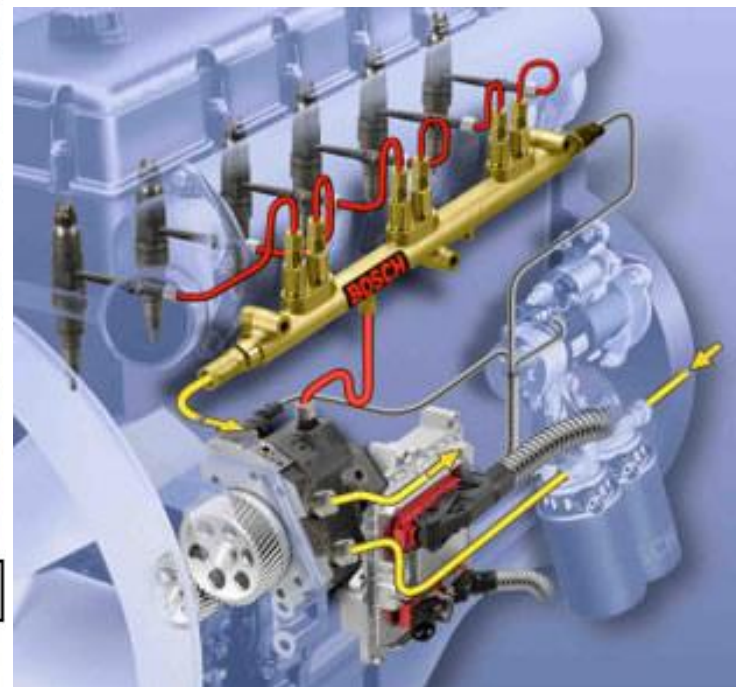
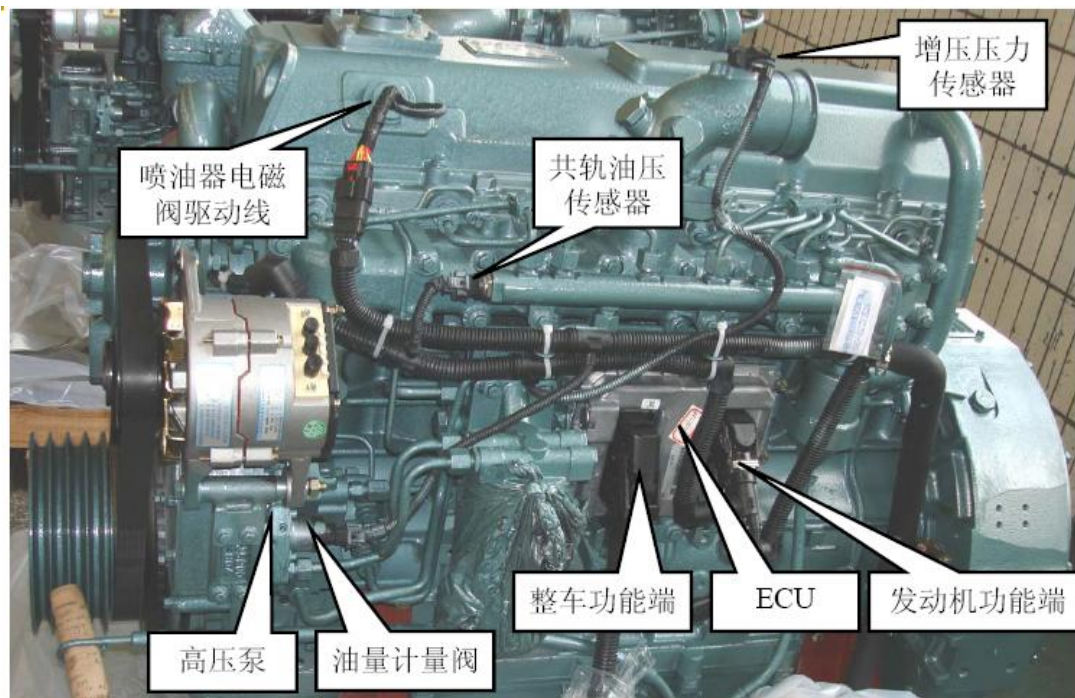
按功能分有两极调速器、全速调速器、定速调速器和综合调速器。

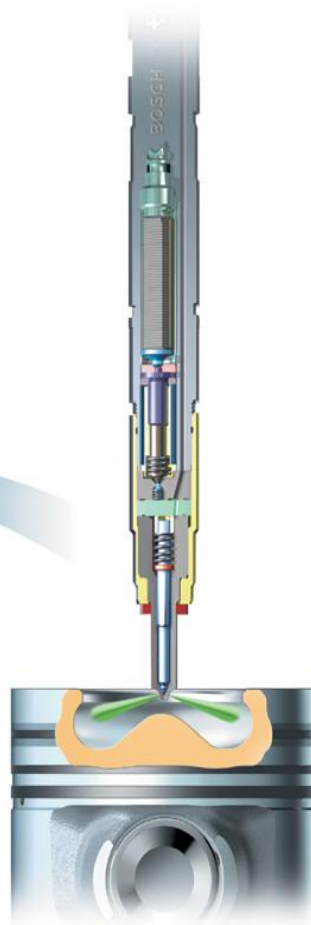
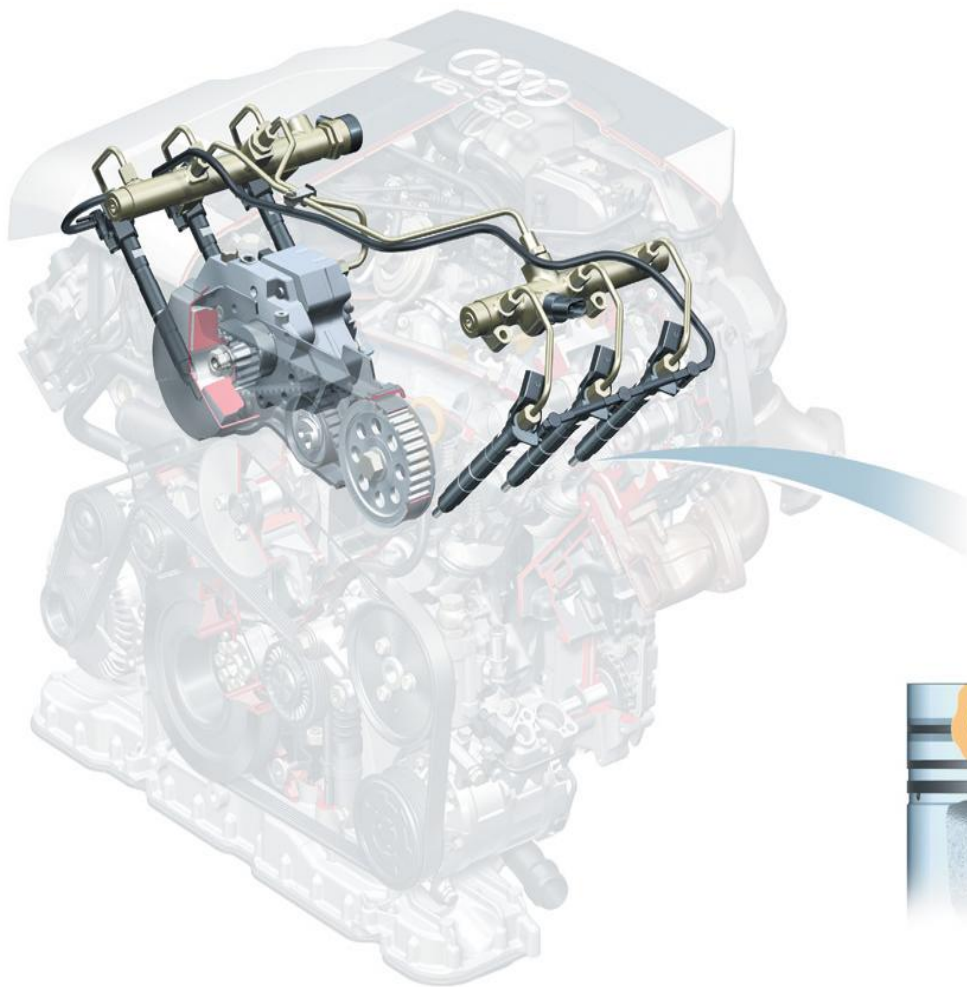
- ✓ In terms of sensor: pneumatic governor, centrifugal governor, complex governor

按转速传感分有气动式调速器、机械离心式调速器和复合式调速器。

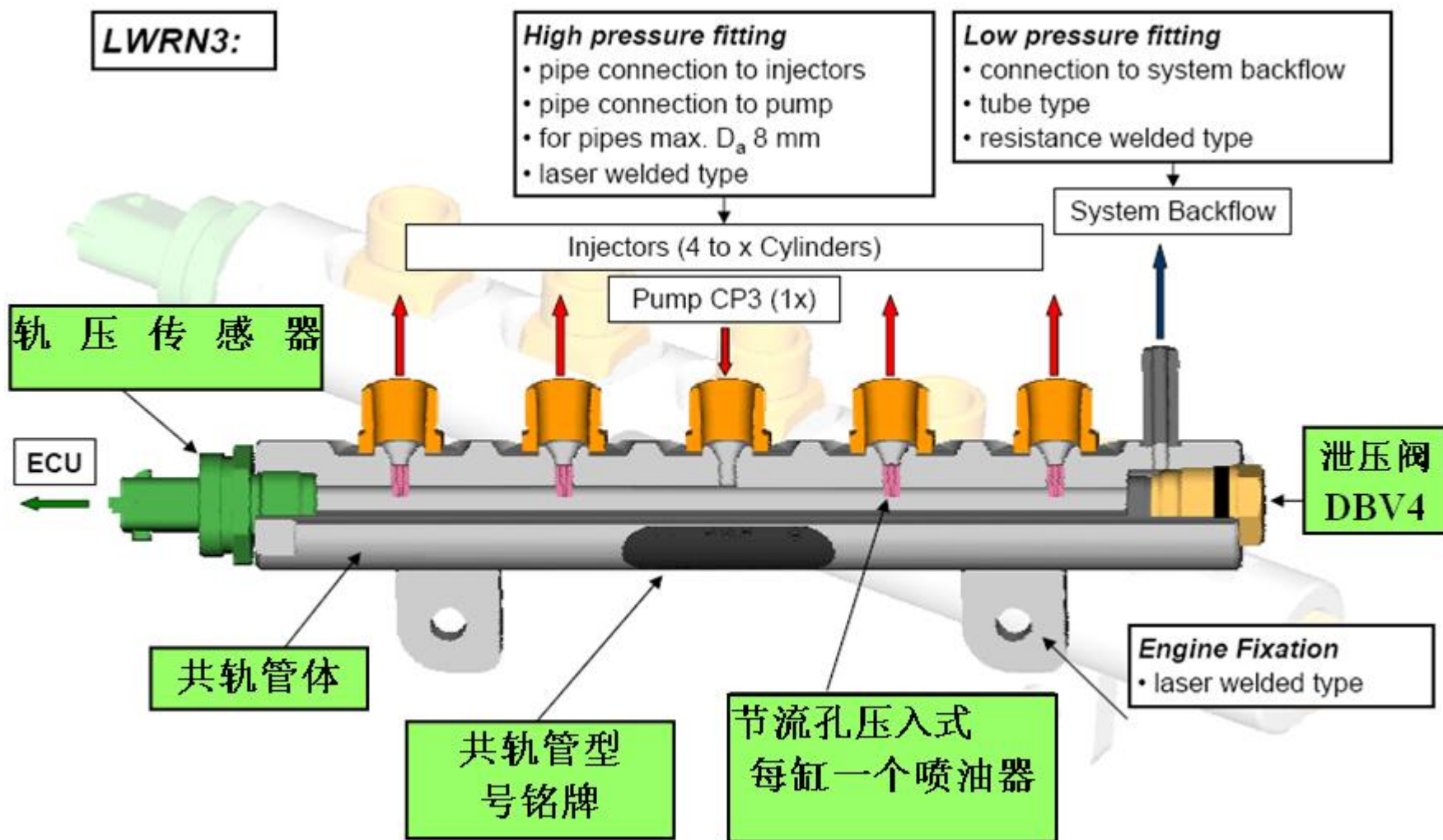


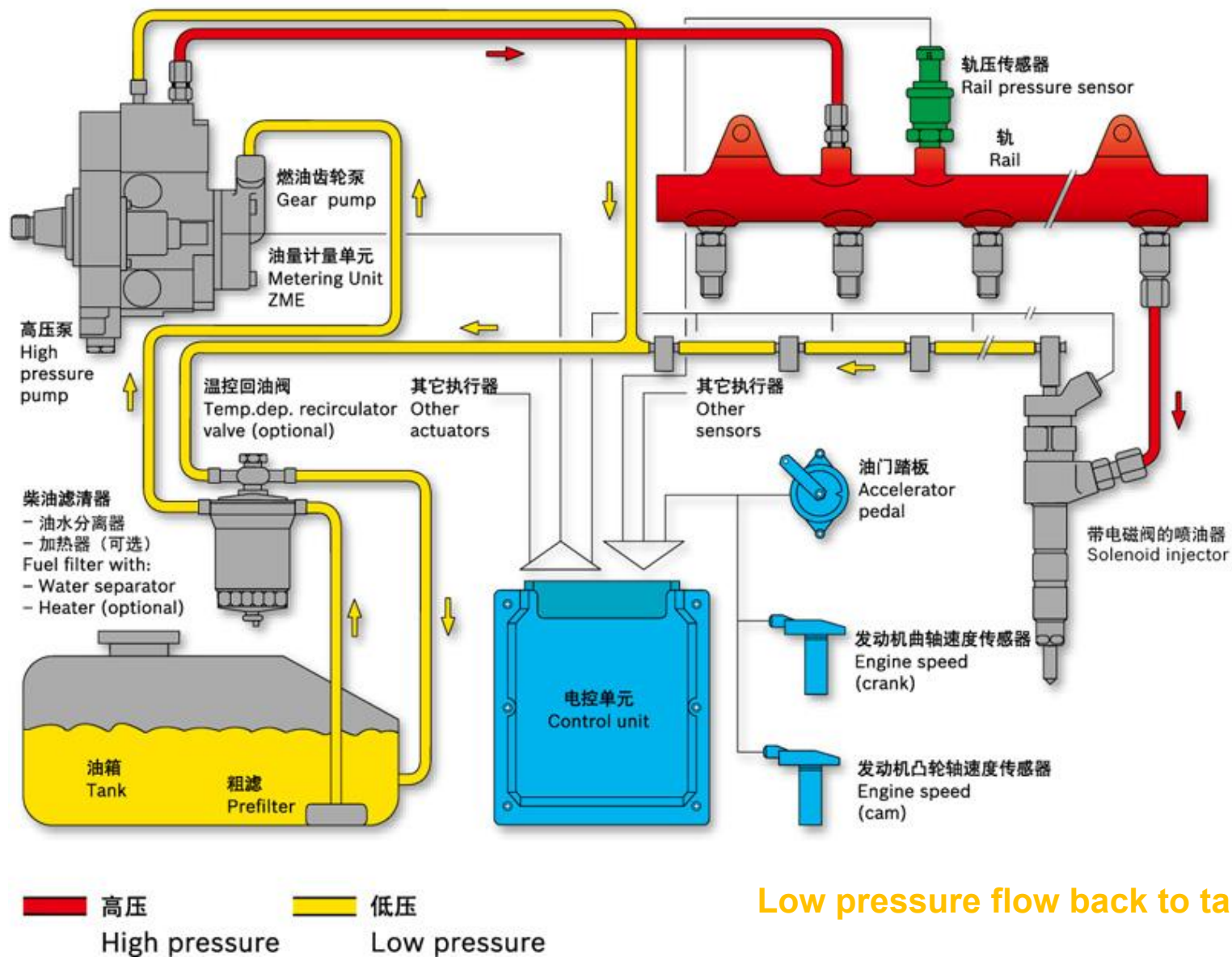
5.7 Common Rail Injection System (共轨系统)



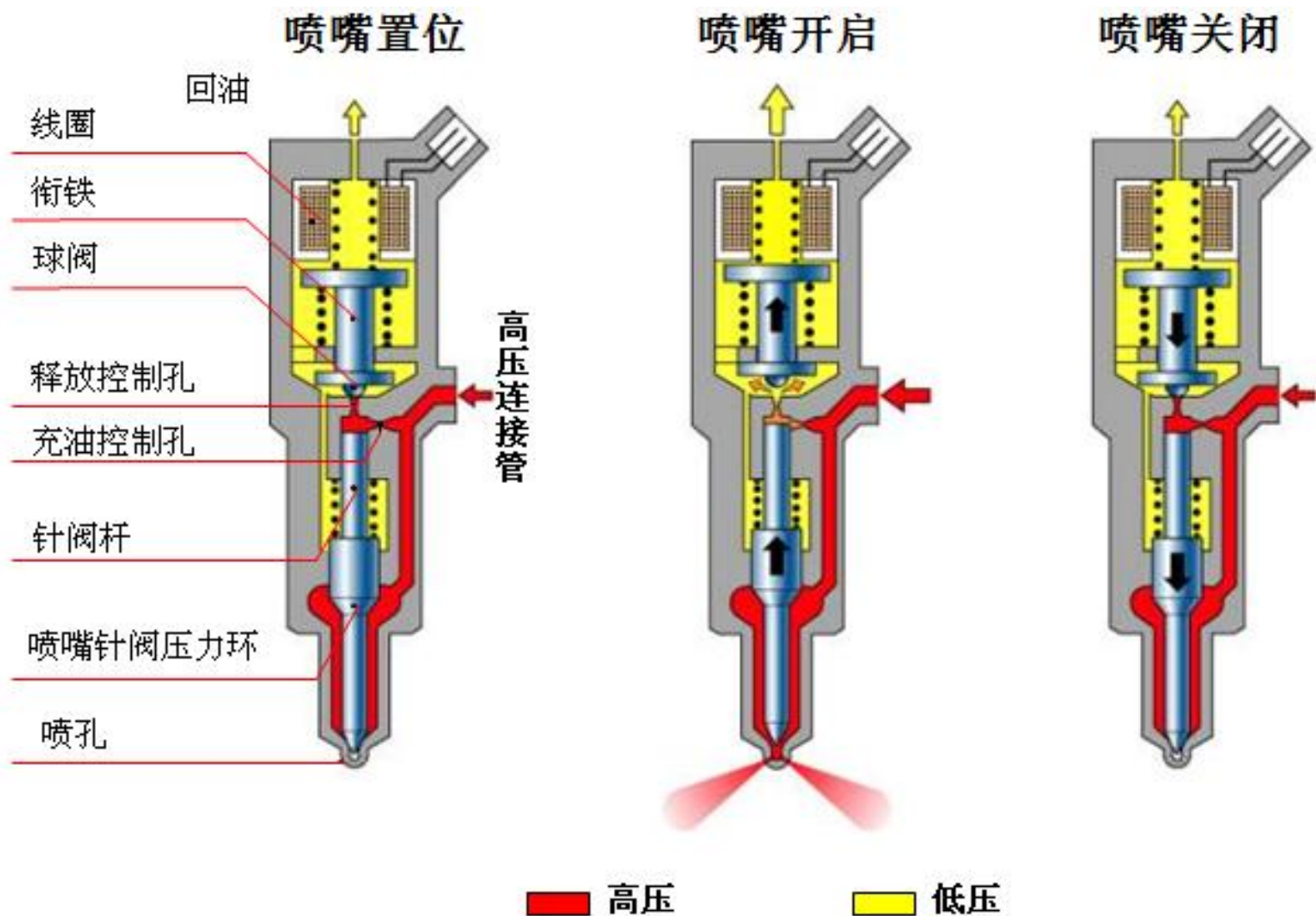


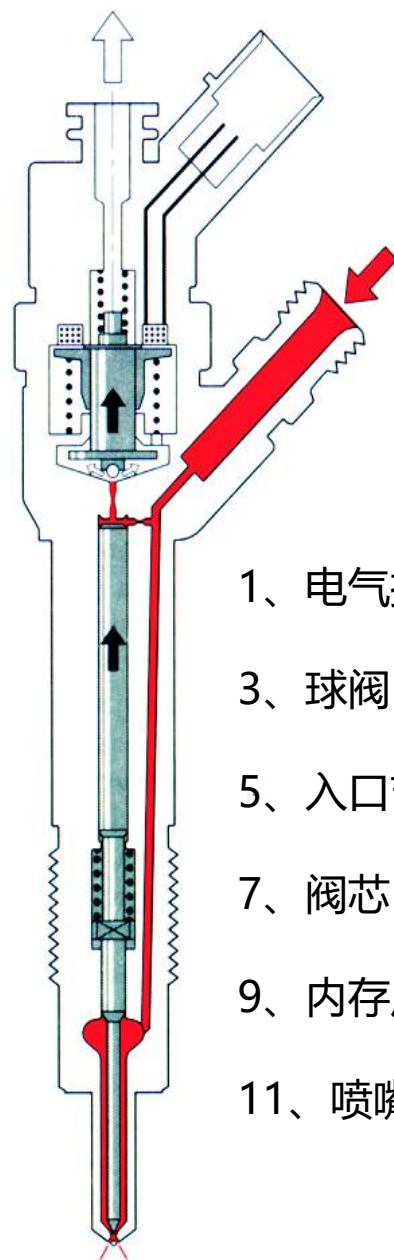
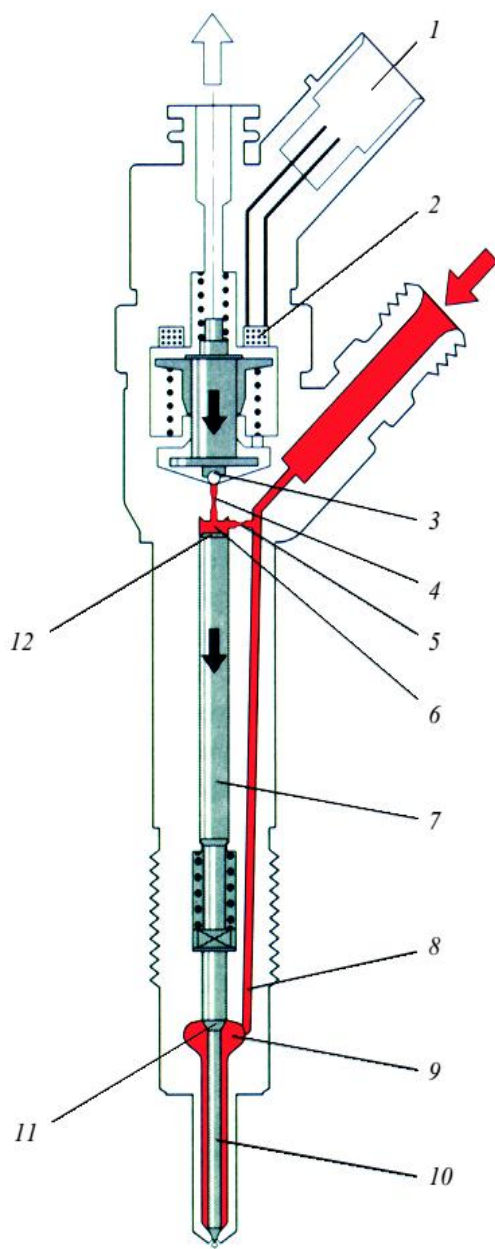






Low pressure flow back to tank





1、电气接口

2、电磁铁

3、球阀

4、出口节流

5、入口节流

6、内存压力

7、阀芯

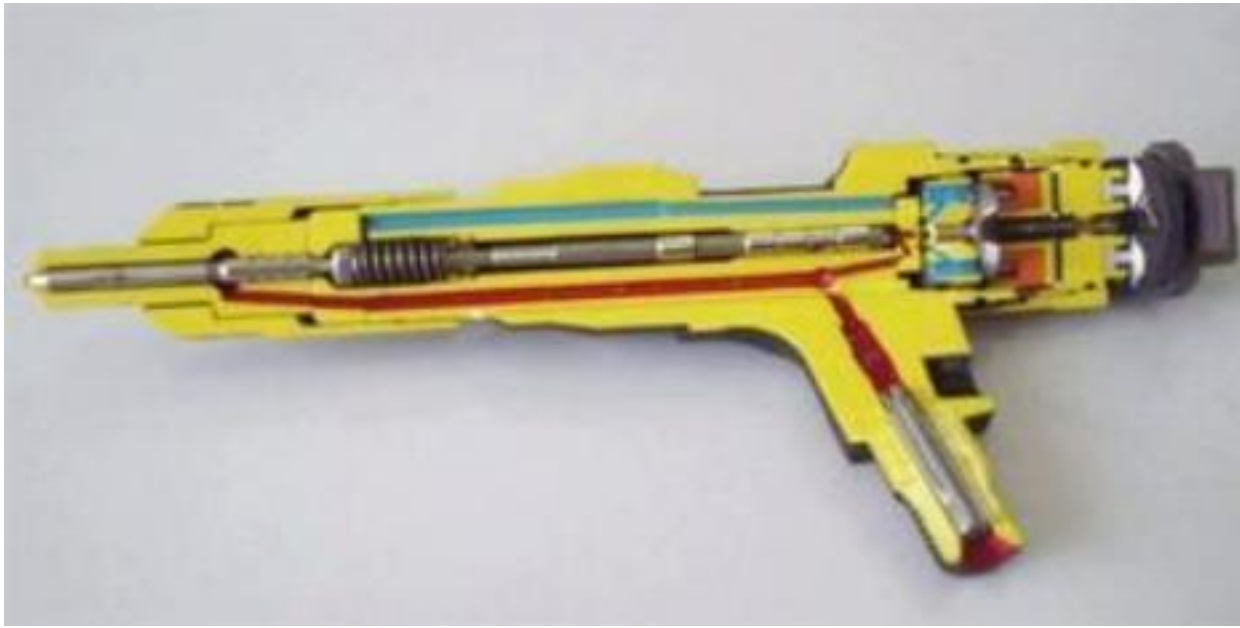
8、喷嘴入口通道

9、内存压力

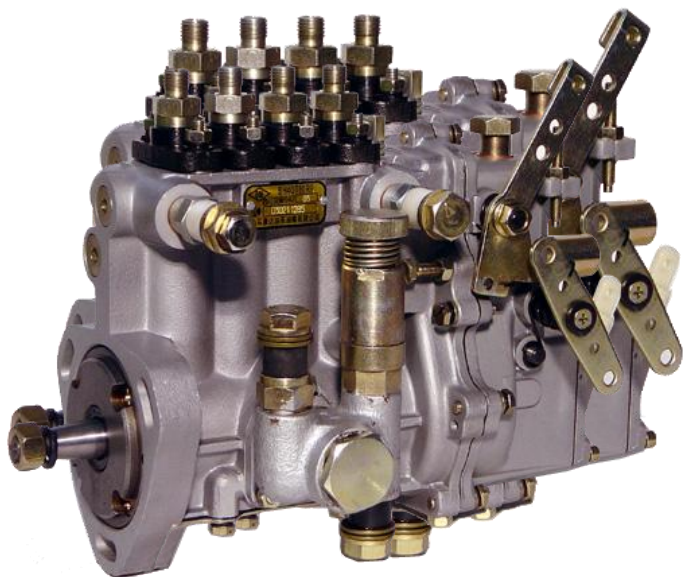
10、喷嘴针阀

11、喷嘴针阀锥形表面

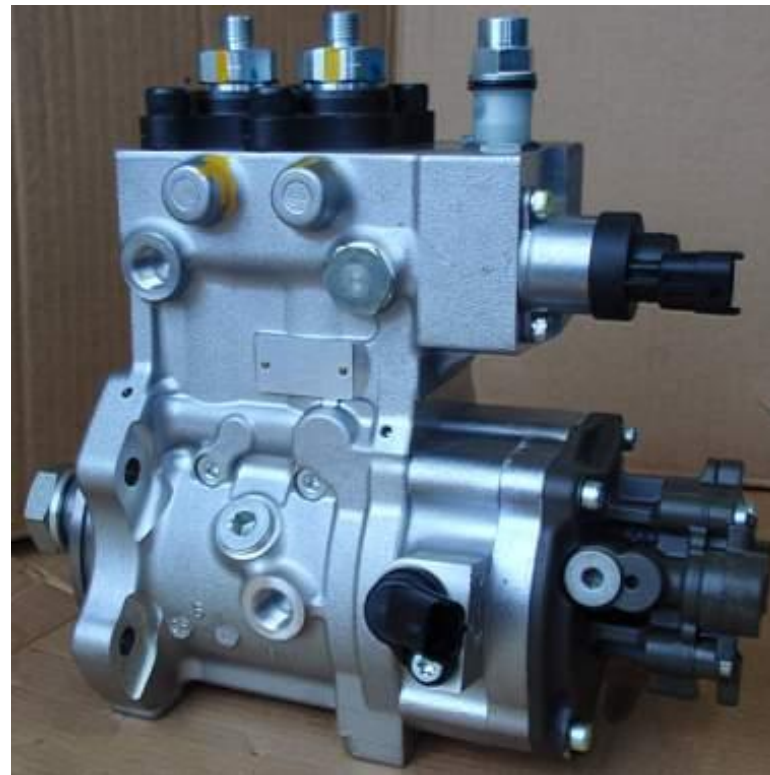
12、压力控制活塞表面



- **Injector**
- Satisfy the automotive requirements regarding temperature and vibration
- Quick response, especially for multi-injection
- **Classification**
- Solenoid injector
 - ✓ higher electrical voltage
 - ✓ higher working temperature
- Piezoelectric injector(压电式)
 - ✓ Quick response
 - ✓ Lower working temperature

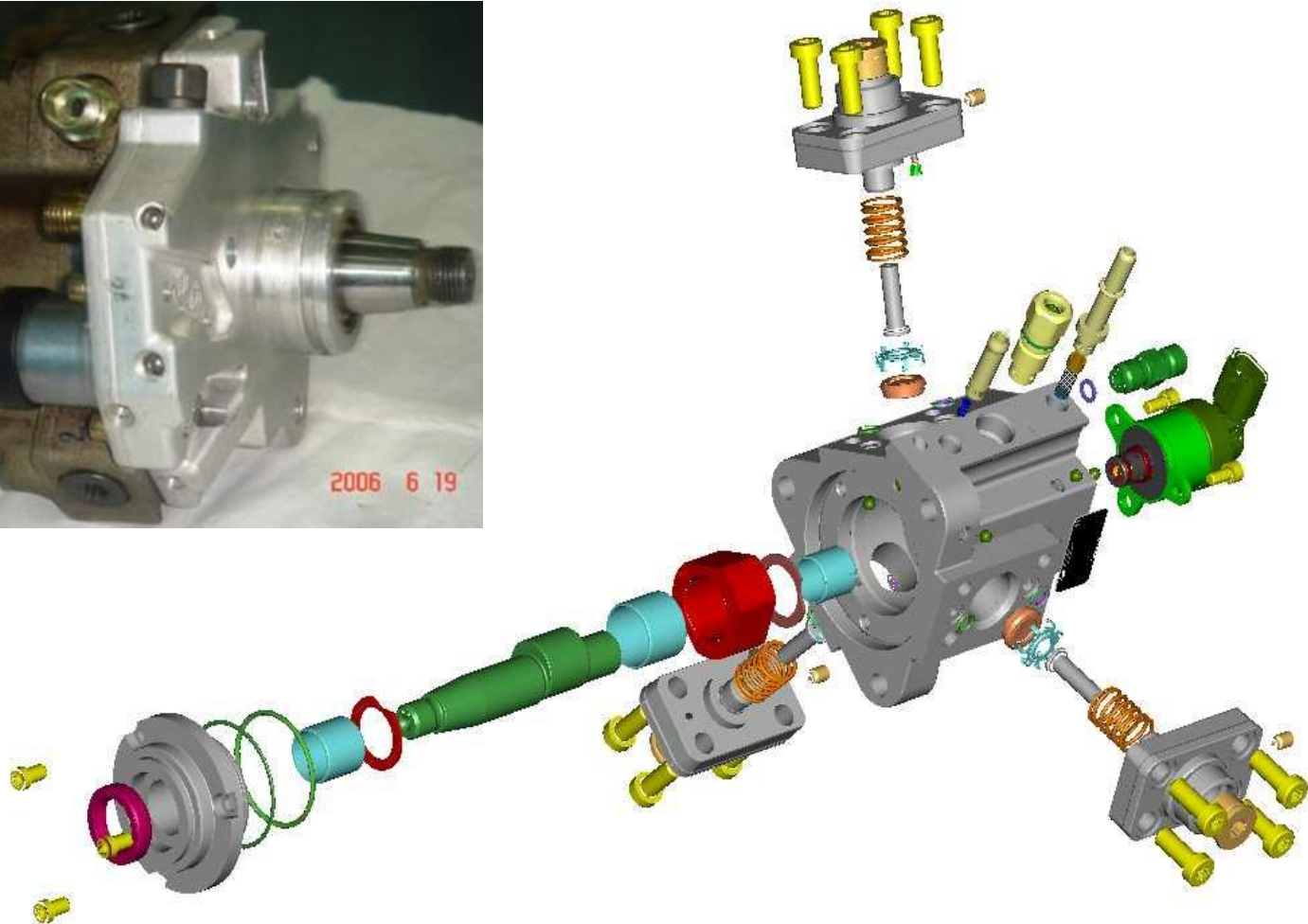


常规机械泵

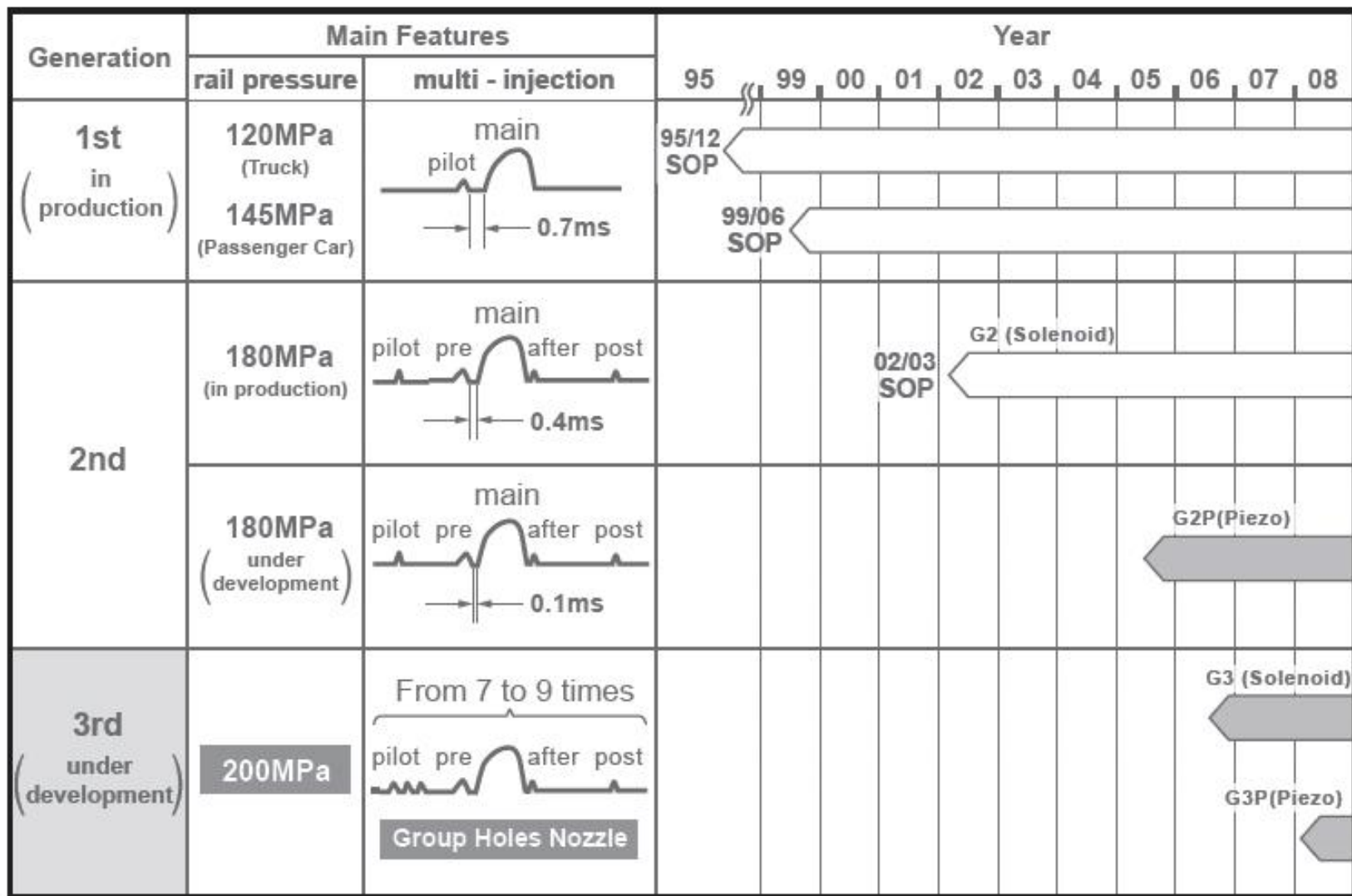


直列双柱塞高压油泵

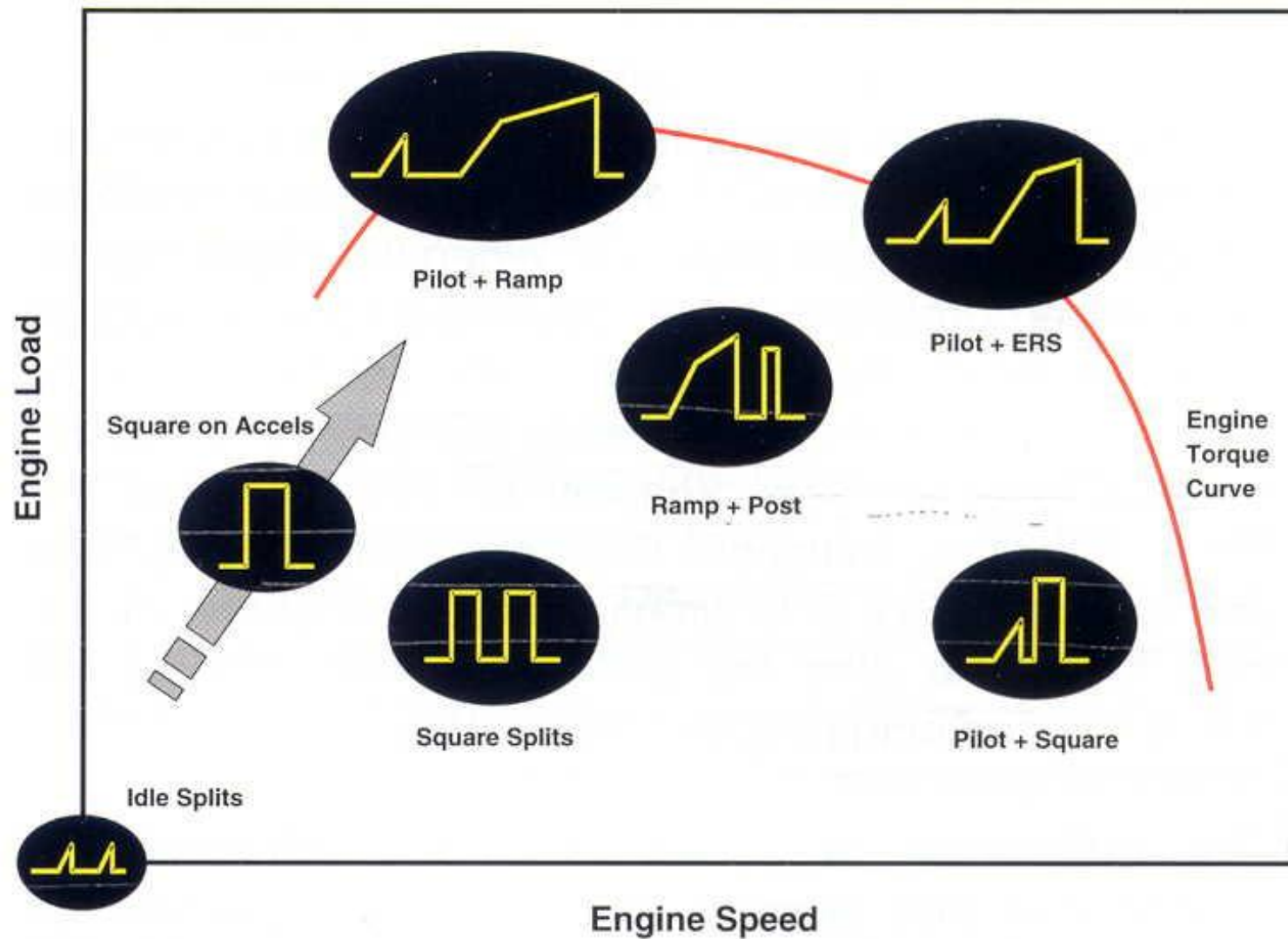
Less plungers

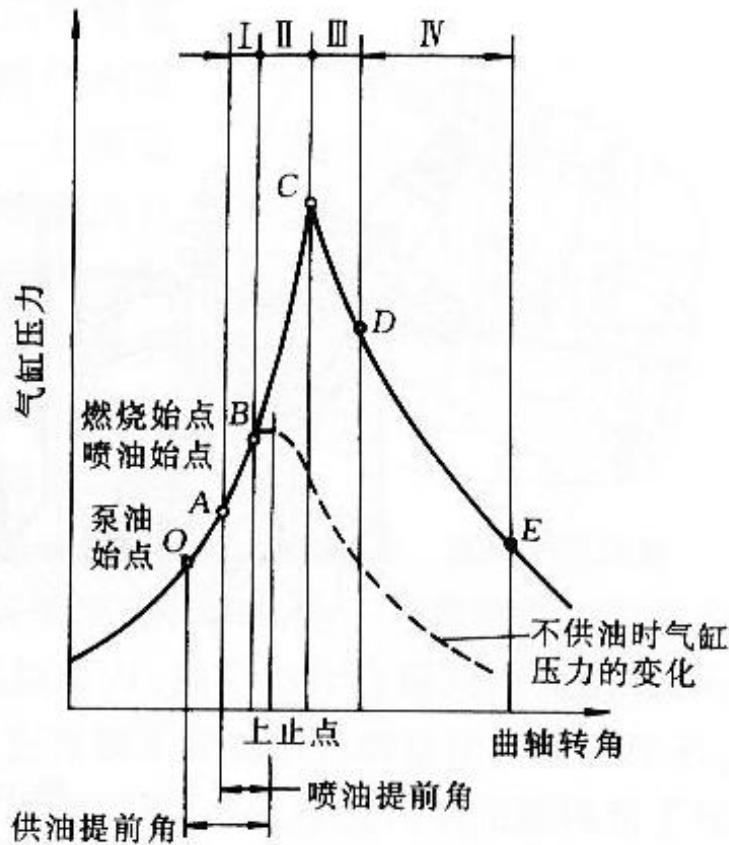


3-缸径向柱塞高压油泵



DENSO CR system road map





柴油机气缸压力与曲轴转角的关系

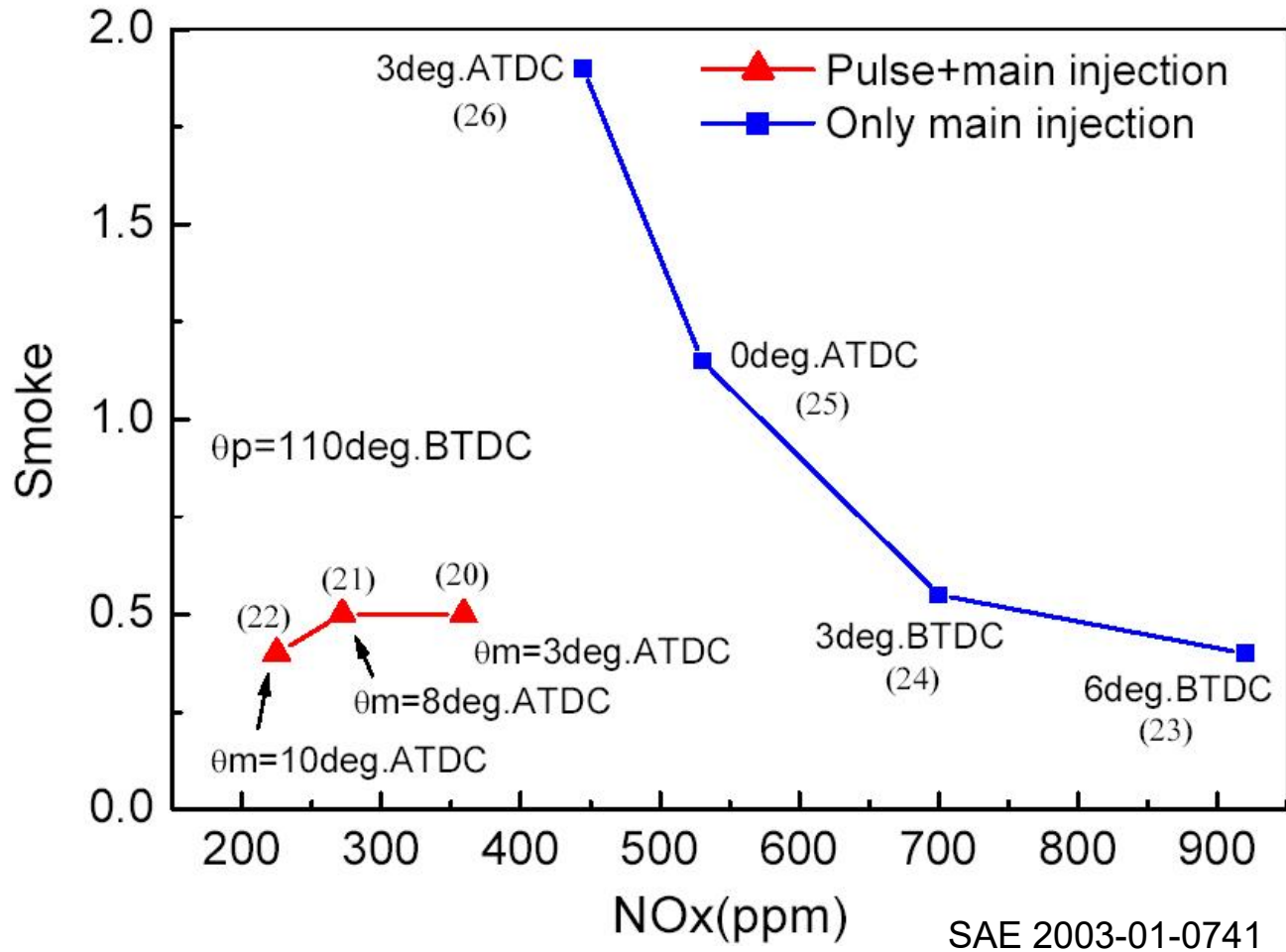
I—滞燃期； II—速燃期； III—缓燃期； IV—后燃期

Combustion process can be divided into four periods:

- **A-B ignition delay period (滞燃期)**
From start of injection to start of combustion (combustion line leaves compression line)
- **B-C rapid combustion period (急燃期)**
From start of combustion to appearance of maximum cylinder pressure
- **C-D Slow combustion period (缓燃期)**
From appearance of maximum cylinder pressure to maximum cylinder temperature
- **D-E After combustion period (后燃期)**

Comparison between Single and Multiple Injections

(IMEP=0.93MPa, case 20~22, $\theta_p=110\text{deg. BTDC}$,
 $P_{cr}=80\text{MPa}$, 1400r/min)



3rd Paper Topic

汽车发动机及零部件的新发展与趋势

The latest developments and trends of car engine and its components

4th Paper Topic

清洁零排放汽车动力系统的现状与趋势

The current achievements and future and trends of clean and zero-emission automobile powertrains